

Section 1: Identification

- **Product Name:** Phosphoric Acid and 2% Sodium Chloride Solution
- **Synonyms:** Phosphoric Acid and Salt Solution
- **Product Code:** Nital Salt
- **Manufacturer:**
PACE Technologies
3601 E. 34th St.
Tucson, AZ 85713
+1-520-882-6598
- **Emergency Phone Number:**
CHEMTREC 800-424-9300 (US) Day or night
Customer No. 16568

Section 2: Hazard Identification

Classification of the Substance or Mixture (GHS Classification)

- **Phosphoric Acid (H₃PO₄):**
 - Corrosive to Metals: Category 1
 - Skin Corrosion/Irritation: Category 1B
 - Serious Eye Damage/Irritation: Category 1
- **2% Sodium Chloride (NaCl):** Not hazardous at this concentration

Label Elements

- **Pictograms:**



- **Signal Word: Danger**
- **Hazard Statements:**
 - **Phosphoric Acid:**

- H290: May be corrosive to metals.
 - H314: Causes severe skin burns and eye damage.
 - **2% Sodium Chloride:** Not hazardous at this concentration.
- **Precautionary Statements:**
 - **Prevention:**
 - P234: Keep only in original packaging.
 - P260: Do not breathe dust/fume/gas/mist/vapors/spray.
 - P264: Wash hands thoroughly after handling.
 - P280: Wear protective gloves/protective clothing/eye protection/face protection.
 - **Response:**
 - P301 + P330 + P331: IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
 - P303 + P361 + P353: IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower.
 - P305 + P351 + P338: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses if present and easy to do – continue rinsing.
 - **Storage:**
 - P405: Store locked up.
 - **Disposal:**
 - P501: Dispose of contents/container to an approved waste disposal facility in accordance with local, regional, national, and international regulations.

Section 3: Composition/Information on Ingredients

Substance	CAS Number	Concentration (% w/w)
Phosphoric Acid	7664-38-2	98-99%
Sodium Chloride (NaCl)	7647-14-5	1-2%

Substance	CAS Number	Concentration (% w/w)
Water	7732-18-5	Balance

Section 4: First-Aid Measures

- **Inhalation:** Move to fresh air. If breathing is difficult, administer oxygen. Seek medical attention if symptoms occur.
 - **Skin Contact:** Immediately flush with plenty of water for at least 15 minutes. Remove contaminated clothing and shoes. Seek medical attention for burns or persistent irritation.
 - **Eye Contact:** Rinse cautiously with water for at least 15 minutes. Remove contact lenses if possible. Seek immediate medical attention.
 - **Ingestion:** Do NOT induce vomiting. Rinse mouth with water. Seek medical attention immediately.
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Section 5: Fire-Fighting Measures

- **Suitable Extinguishing Media:** Use water spray, dry chemical, carbon dioxide, or foam. Do not use water jets directly on the acid.
 - **Special Hazards Arising from the Substance or Mixture:** Phosphoric acid may decompose at high temperatures, releasing toxic and corrosive fumes including phosphorus oxides.
 - **Protective Equipment for Firefighters:** Use self-contained breathing apparatus and full protective gear.
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Section 6: Accidental Release Measures

- **Personal Precautions:** Evacuate the area. Use appropriate personal protective equipment (PPE). Ensure proper ventilation.
 - **Environmental Precautions:** Prevent spillage from entering drains, waterways, or soil.
 - **Methods for Containment and Cleanup:** Neutralize small spills with a dilute sodium carbonate (Na_2CO_3) solution. Absorb with inert material, and dispose of in accordance with local regulations.
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Section 7: Handling and Storage

- **Precautions for Safe Handling:** Avoid contact with skin, eyes, and clothing. Do not inhale vapors. Handle in a well-ventilated area.
 - **Conditions for Safe Storage:** Store in corrosion-resistant containers in a cool, well-ventilated area away from incompatible materials such as strong bases and metals.
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Section 8: Exposure Controls/Personal Protection

Control Parameters

- **Phosphoric Acid:**
 - OSHA PEL: 1 mg/m³
 - ACGIH TLV: 1 mg/m³ (TWA); 3 mg/m³ (STEL)

Personal Protective Equipment (PPE)

- **Respiratory Protection:** If airborne concentrations exceed the permissible exposure limit, use a NIOSH-approved respirator.
 - **Hand Protection:** Wear acid-resistant gloves (e.g., nitrile, butyl rubber).
 - **Eye Protection:** Wear chemical safety goggles and/or a full face shield.
 - **Skin Protection:** Use acid-resistant clothing as needed to protect from spills and splashes.
 - **Engineering Controls:** Ensure adequate ventilation, especially in confined areas.
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Section 9: Physical and Chemical Properties

- **Appearance:** Clear, colorless liquid
- **Odor:** Odorless
- **pH:** Acidic (dependent on phosphoric acid concentration)
- **Melting/Freezing Point:** Approx. 0°C (for water-based solution)
- **Boiling Point:** Approx. 100°C
- **Flash Point:** Not applicable
- **Solubility:** Completely soluble in water

- **Vapor Pressure:** Similar to water (for dilute solutions)
 - **Density:** Varies with concentration
 - **Viscosity:** Similar to water (for dilute solutions)
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Section 10: Stability and Reactivity

- **Reactivity:** Phosphoric acid reacts with strong bases, metals, and oxidizing agents.
 - **Chemical Stability:** Stable under normal conditions.
 - **Conditions to Avoid:** Avoid exposure to heat and incompatible materials.
 - **Incompatible Materials:** Strong bases, metals (may release hydrogen gas), strong oxidizers.
 - **Hazardous Decomposition Products:** Phosphorus oxides.
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Section 11: Toxicological Information

- **Acute Toxicity:**
 - **Phosphoric Acid:** May cause burns to skin and eyes. Harmful if ingested or inhaled.
 - **Sodium Chloride:** Non-hazardous at low concentrations.
 - **Skin Corrosion/Irritation:** Phosphoric acid causes severe burns.
 - **Serious Eye Damage/Irritation:** Causes serious eye damage (phosphoric acid).
 - **Carcinogenicity:** Not classified as a carcinogen.
 - **Reproductive Toxicity:** No known significant effects.
 - **Aspiration Hazard:** Not applicable.
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Section 12: Ecological Information

- **Ecotoxicity:** Phosphoric acid can be harmful to aquatic life in large quantities.
- **Persistence and Degradability:** Phosphoric acid is readily neutralized in the environment.
- **Bioaccumulative Potential:** Phosphoric acid does not bioaccumulate.

- **Mobility in Soil:** High mobility due to water solubility.
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Section 13: Disposal Considerations

- **Waste Disposal:** Dispose of waste in accordance with local, regional, national, and international regulations. Neutralize the solution before disposal.
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Section 14: Transport Information

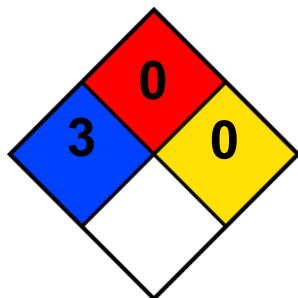
- **UN Number:** UN1805
 - **Proper Shipping Name:** Phosphoric Acid Solution
 - **Hazard Class:** 8 (Corrosive)
 - **Packing Group:** III
 - **Environmental Hazards:** None known.
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Section 15: Regulatory Information

- **OSHA:** Hazardous according to OSHA Hazard Communication Standard.
 - **SARA 313:** Not subject to reporting.
 - **TSCA:** All components are listed on the TSCA Inventory.
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Section 16: Other Information

16.1 NFPA 704



- **Health (3):** Phosphoric acid is corrosive and can cause severe skin burns, eye damage, and respiratory irritation. A rating of 3 indicates serious injury could result from exposure.
- **Flammability (0):** Phosphoric acid and sodium chloride solutions are non-flammable.
- **Reactivity (0):** Phosphoric acid in dilute form (like in this mixture) is stable under normal

conditions. A rating of 0 means that the mixture is generally stable and poses no significant reactivity hazard.

- **Special Hazards (none):** No special hazard symbols are required for this mixture.

Product Use:

Laboratory Reagent.

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DATE REVISED: 7/22/2025 DZ
